

## CY 553: Organic Inorganic Spectroscopy

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|---------------|-------------------------------------------------|
| Course Code   | CY 553                                          |
| Course Name   | Organic Inorganic Spectroscopy                  |
| L-T-P-C       | 3-0-0-3                                         |
| Prerequisites | B.Sc in Sciences with Chemistry as main subject |
| Intended for  | PG                                              |
| Distribution  | Elective for Masters in Chemistry               |
| Approval      | 9th Senate (May 2016)                           |

### Course Contents

- **Recapitulation of UV, IR, NMR and Mass spectroscopy** (In this module the elaborated syllabus has been included for a better idea of the topics to be covered)  
UV includes: Color and light absorption, the cLecturesomophore concept. Theory of electronic spectroscopy, orbitals involved and electronic transitions, effect of solvent and conjugation on max. Woodward Fieser rules e.t.c.  
IR includes: Molecular vibrations and modes of vibrations. Factors influencing vibrational frequencies vibrational coupling, hydrogen bonding, conjugation, inductive, mesomeric (resonance), field effects and bond angles, application to identify functional groups.  
NMR includes: Nuclear spins resonance, chemical shift and its measurement. Relaxation processes, factors influencing chemical shift. Shielding, deshielding and anisotropic effects. Effect of restricted rotation, concentration temperature and hydrogen bonding. Spin coupling (simple and complex), mechanism of coupling.  
Mass includes: Introduction, Mass spectrum and Metastable ion peak, Determination of molecular formula and recognition of molecular ion peak and the Nitrogen rule, Molecular formula and index of Hydrogen deficiency, General rules of fragmentation and the McLafferty rearrangement, Fragmentations associated with functional groups. Structure elucidation of organic compounds by joint applications of IR, NMR and Mass spectroscopy. [12 Lectures]
- **Circular DicLecturesoism** (This is a new module introduced on the suggestions of reviewers)  
Theory of polarized light, optical activity and optically active molecules, Cotton effects, CD, Octet Rule, Experimental Techniques, Applications. [6 Lectures]
- **Nuclear Quadrupole Resonance Spectroscopy**  
Introduction, energies of quardrupole transitions, effect of magnetic field on the spectra, relationship between electric field gradient and molecular structure, applications, interpretations of structural information from NQR spectra. [7 Lectures]
- **Mossbauer Spectroscopy**  
The Mossbauer effect, the Mossbauer active nuclei.. The chemical isomer shift. The quadrupole splitting, magnetic hyperfine interactions. Mossbauer spectroscopy applied to study nature of chemical bond, structural determination and analytical applications. [8 Lectures]
- **Electron Paramagnetic Resonance Spectroscopy**  
Introduction, Principles, Presentation of the spectrum, hyperfine splitting in isotropic systems involving more than one nucleus, Contributions to the hyperfine coupling constant in isotropic systems. Anisotropic Effects: Anisotropy in the g value, EPR of triplet states, nuclear quadrupole interaction, line widths, EPR applications. [9 Lectures]

### Text and Reference Books

1. R. V. Parish, **NMR, NQR, EPR and Mossbauer Spectroscopy In Inorganic Chemistry**, Elis Horwood Limited.
2. Robert M. Silverstein, Francis X. Webster, David J. Kiemle, David L. Bryce, **Spectrometric Identification of Organic Compound**, John Wiley and Sons.
3. William Kemp, **Organic Spectroscopy**, Macmillan.
4. Russell S. Drago, **Physical Methods in Inorganic Chemistry**, Van Nostrand Reinhold.
5. Skoog, West and Holler, **Fundamentals of Analytical Chemistry**, Saunders College publishing.
6. Field, Sternhell & Kalman, **Organic structures from Spectra**, John Wiley and Sons.